

Week 8 Group Assignment- Project Group #3

Problem 2

Assume a program begins with the statement:

```
allData = {  
'US': {'pop': 325.7, 'gdp': 19.39, 'ccy': 'USD', 'fx': 1.0},  
'CA': {'pop': 36.5, 'gdp': 1.65, 'ccy': 'CAD', 'fx': 1.35},  
'MX': {'pop': 129.2, 'gdp': 1.15, 'ccy': 'MXN', 'fx': 19.68}  
}
```

Write subsequent statements that (in a loop) prompt the user for a country code (US, CA, or MX), then prompt for a measure name (pop, gdp, ccy, or fx), then look up in the above dictionary for the corresponding value, and display that value.

Example run:

```
Please enter a country code: CA  
Please enter a statistic: pop  
CA pop = 36.5
```

Your program should handle the cases where a user-entered country code or measure name is not found.

Problem 3: Final Project

Take this time to also start brainstorming ideas for your final project while in class. A list of ideas was provided by the instructor.

Your task:

write a summary of what you have done today regarding your project.

Problem 1:

```
# Birthday Dictionary (code)  
birthdays = {}  
choice = 0  
  
while choice != 5:  
    # menu  
    print("\n Birthday Menu, please select an option below. ")  
    print("1. Look up a birthday")  
    print("2. Add a new birthday")  
    print("3. Change a birthday")  
    print("4. Delete a birthday")  
    print("5. Quit the program")  
    choice = int(input("Enter your choice: "))  
  
    if choice == 1:  
        # look up birthday  
        name = input("Enter the name to look up: ")  
        if name in birthdays:  
            print(name + "'s birthday is " + birthdays[name])  
        else:  
            print(name + " is not in the dictionary.")
```

```

elif choice == 2:
    # add new birthday
    name = input("Enter the name to add: ")
    if name in birthdays:
        print(name + " already exists. Use option 3 to change their
birthday.")
    else:
        birthday = input("Enter " + name + "'s birthday: ")
        birthdays[name] = birthday
        print(name + "'s birthday has been added.")
elif choice == 3:
    # change a birthday
    name = input("Enter the name whose birthday you want to change: ")
    if name in birthdays:
        birthday = input("Enter the new birthday for " + name + ": ")
        birthdays[name] = birthday
        print(name + "'s birthday has been updated.")
    else:
        print(name + " is not in the dictionary.")
elif choice == 4:
    # delete a birthday
    name = input("Enter the name to delete: ")
    if name in birthdays:
        del birthdays[name]
        print(name + " has been deleted from the dictionary.")
    else:
        print(name + " is not in the dictionary.")
elif choice == 5:
    # quit program
    print("Program closing, Goodbye!")
else:
    print("Invalid choice. Please enter a number between 1 and 5.")

```

Problem #2:

```

#problem 2
allData = {
    'US': {'pop': 325.7, 'gdp': 19.39, 'ccy': 'USD', 'fx': 1.0},
    'CA': {'pop': 36.5, 'gdp': 1.65, 'ccy': 'CAD', 'fx': 1.35},
    'MX': {'pop': 129.2, 'gdp': 1.15, 'ccy': 'MXN', 'fx': 19.68}
}
#loop so program keeps running
while True:
    country = input("Please enter a country code: ")
    #ask user for statistic
    stat = input("please enter a statistic")
    # check if the country exists
    if country in allData:
        # check if the statistic exists for that country
        if stat in allData[country]:
            # print the value from the dictionary
            print(country, stat, "=", allData[country][stat])
        else:
            # message if statistic not found
            print("Statistic not found")
    else:
        # message if country not found
        print("Country not found")

```

Problem #3:

-Game (Educational Wordle) 4-6th grade science class tool Have the user choose a grade (per user choice have API pull set of words from that grade and science subject)

Ex:

_____(Plant)

Hint: An organism that survives through photosynthesis After 3 or 4 wrong attempts, give a hint after the 3rd failed attempt and a letter after the 4th attempt.

Purpose: This educational game serves as a new method to educate students between 4th to 6th grade on their current level of science and to provide a fun opportunity to learn. The rules are as follows. First, the user will input the grade level into the software. Then there will be a definition of a scientific term used in the users' grade level. Then the user will have 6 opportunities to guess the correct 5 letter word for the definition. If the user inputs a letter not used in the defined word, then it will be colored RED. If the letter is correct but in the incorrect spot, the letter will be YELLOW. If the letter is correct and in the correct spot, the letter will become GREEN. Once the user has found the correct letters and formed the correct word, all the letters will turn green and the user will win the game. The game will automatically end after 5 failed attempts. Regardless of the results, the correct word and definition will be displayed to help the user understand the concept despite the result. Through the use of API, we can pull 5 letter terms and definitions based on the users' input of their grade level. This will allow players to learn and discover multiple terms to better educate the user, while also providing a fun popular game from the mainstream. We believe this checks all the boxes of a beneficial and fun use of coding.